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Observations and Orbital Analysis of the High-Amplitude Delta Scuti Star SZ Lyncis: The Unusual Orbital Precession

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Abstract

We determined forty-two new times of light maximum from our photometry observations and WASP project, and collected all times of light maximum observed between 1961 and 2013 in order to calculate the orbital elements of the SZ Lyncis system and the secular change of the pulsation period with the classical $O-C$ method. We confirmed the decrease of the longitude of the periastron passage with a rate of $(-1\overset{\circ}{.}15 \pm 0\overset{\circ}{.}25)\,\mathrm{yr}^{-1}$, and discussed the causative mechanism. The results show that the precession of the star's orbit might be due to a close binary system, which means that the companion of SZ Lyncis is actually a binary system. We used the Hipparcos Intermediate Astrometric Data to obtain the complete orbital elements of the SZ Lyncis system, and found that the inclination, i, and parallax, π_t, are $39\overset{\circ}{.}5 \pm 17\overset{\circ}{.}7$ and 2.61 ± 0.98 mas (corresponds to 380 ± 140 pc), respectively. We reanalyzed the mean radial velocities of SZ Lyncis given by Bardin and Imbert (1984), and noticed a weak variation existing in the residuals from a single-Keplerian fit. We suggest that more detailed high-precision spectroscopic observations are definitely needed in the future to check this short periodic change.

Key words: stars: variables: delta Scuti — stars: individual: SZ Lyncis — techniques: photometric

1. Introduction

SZ Lyncis ($\alpha_{2000} = 08^{\mathrm{h}}09^{\mathrm{m}}35\overset{\mathrm{s}}{.}7$, $\delta_{2000} = +44^{\circ}28'17\overset{''}{.}6$, $\langle V \rangle = 9.40$, $\Delta V = 0.64$, Sp: A7-F2) is a high-amplitude δ Scuti-type star (HADS), a member of a single-lined spectroscopic binary. The δ Scuti-type stars are pulsators situated in the classical Cepheid instability strip on the main sequence, or moving from the main sequence to the giant branch (Breger 2000). The pulsation period range is between 0.02 and 0.25 d, and the mass range is between 1.5 and 2.5 $M_{\odot}$. The oscillation modes are radial and nonradial at low-degree p modes. High-amplitude δ Scuti stars are δ Scuti stars with V amplitudes $\geq 0\overset{\mathrm{m}}{.}30$, and Population II members of the group are also known as SX Phoenicis stars. Many δ Scuti stars have been found in double and multiple systems, including very wide binaries, spectroscopic binaries and eclipsing binaries (Lampens & Boffin 2000; Zhou 2010; for HADSs, Derekas et al. 2009). Because calculations of the orbits of binaries can determine the stellar parameters (mass, radius, and density) of the component stars, investigating a δ Scuti star when it is in a stellar system can give more important information than a single-field one.

SZ Lyncis was reported to be a variable star by Hoffmeister (1949a, 1949b), and light curves were first determined by Schneller (1961). van Genderen (1967) first pointed out that the $(O-C)$s produced by a linear ephemeris appeared to follow a sine-curve relation with a period of 3.091 ± 0.051 yr; this value seems to change with time (Szeidl 1983; Paparó et al. 1988). Barnes and Moffett (1975) interpreted that the periodicity in an $O-C$ diagram was caused by the light-travel time effect, leading to the prediction that SZ Lyncis is a single-lined spectroscopic binary. Bardin and Imbert (1981, 1984) observed SZ Lyncis with the CORAVEL photoelectric spectrometer from 1979 to 1983, measured twenty-one mean radial-velocity values, and confirmed the binary hypothesis. Paparó et al. (1988) considered SZ Lyncis to be orbiting in the binary system of elliptical orbit, analyzed all of the times of the light maximum from 1961 until 1988 with the $O-C$ method, and obtained the values of the pulsation and orbital elements. They noticed a slight deviation between the photometric and spectroscopic solution in the longitude of the periastron passage, ω. Recent work has been done by Gazeas et al. (2004) as well as Gazeas and Niarchos (2005), but different values of ω were given by them.

The WASP (wide-angle search for planets) project is a ground-based exoplanet transit survey that has been automatically taking wide field images since 2004 (Butters et al. 2010). Due to the wide field of view and long base line, it is also well suited for discovering new stellar variability, or analyzing the short-term period changes of some known variable stars. The quality of the data points for some pulsating stars may not be good enough for spectrum analysis, but sufficient for the $O-C$ analysis. Considering two δ Scuti stars, V1162 Vir and BL Cam have been found to have short-term periodic changes in their $O-C$ diagrams (Arentoft et al. 2001a, 2001b; Fauvaud et al. 2010); similar properties in other

Table 1. The 262 available times of the light maximum of SZ Lyncis determined after the work of Paparó et al. (1988).

JD. Hel. 2400000+	Ref.[*]	JD. Hel. 2400000+	Ref.[*]	JD. Hel. 2400000+	Ref.[*]	JD. Hel. 2400000+	Ref.[*]	JD. Hel. 2400000+	Ref.[*]
48500.0560	1	53397.2734	11	54185.3401	13	54532.47447	WASP	55259.5524	18
51907.4654	2	53397.3933	11	54186.4267	13	54533.43814	WASP	55262.4417[†]	20
51966.5244	3	53409.5689	8	54187.3918	13	54534.40303	WASP	55301.6182	18
51967.3683	3	53409.6874	8	54188.3548	13	54535.36984	WASP	55301.7401	18
52003.4072	3	53416.55775	6	54190.40411	WASP	54536.45279	WASP	55303.4269	21
52032.3351	3	53451.3985[†]	8	54191.3688	13	54537.41626	WASP	55304.5137	22
52053.4294	2	53644.7348	7	54191.4889	13	54539.46564	WASP	55310.4195	22
52202.6467	2	53644.8544	7	54197.3950	13	54544.40729	WASP	55310.5398	22
52209.5192	2	53721.7571	7	54202.3361	13	54544.52745	WASP	55478.8012	18
52263.6389	4	53745.6239	7	54203.4208	13	54545.49175	WASP	55507.4890	22
52308.3564	5	53745.73923[†]	6	54208.6027	7	54547.42052	WASP	55507.6095	22
52321.73578	6	53750.5677	11	54222.5857	7	54553.44820	WASP	55525.8088	18
52343.79563	6	53752.4994[†]	11	54223.4278	13	54554.41195	WASP	55570.6471	23
52374.4127	4	53752.6108[†]	11	54261.6365	7	54562.7261	7	55583.5449	24
52410.3327	4	53752.6160	11	54366.8605	7	54575.38427	WASP	55583.6652	24
52623.55155[†]	6	53758.76187	6	54394.8251	7	54578.0367	YN	55587.6425	24
52623.67342[†]	6	53758.88451	6	54414.8334	7	54583.5797	7	55587.7633	24
52628.5058	5	53759.60604	6	54418.69217	WASP	54758.5988	15	55587.8839	24
52640.54936[†]	6	53759.72692	6	54419.65721	WASP	54768.8438	7	55588.6074	24
52640.66976[†]	6	53759.84713	6	54427.61149	WASP	54781.37899	XL	55588.7273	24
52643.44217[†]	6	53759.96792	6	54436.65147	WASP	54792.8309	7	55588.8483	24
52643.56365[†]	6	53761.5352	7	54436.77142	WASP	54807.6573	7	55588.9688	24
52643.68337[†]	6	53766.2372	11	54438.58043	WASP	54837.5508	7	55591.3810	25
52647.42013[†]	6	53773.5891	7	54438.69925	WASP	54858.5254	16	55591.5000	25
52647.54057[†]	6	53798.41955	6	54439.54408	WASP	54858.6439	16	55591.6210	25
52647.65949[†]	6	53801.5531	7	54479.5591	14	54865.6345	7	55601.6239	23
52648.50325[†]	6	53801.6733	7	54484.6200	7	54865.7558	7	55629.3475	24
52648.62321[†]	6	53808.3052	11	54484.7384	7	54869.3692	17	55639.7115	23
52658.63939	6	53840.6077	7	54491.49422	WASP	54871.4191	17	55654.6578	23
52721.3220	5	53853.6262	7	54497.39774	WASP	54879.6191	18	55838.8355	26
52774.3595	4	53871.5881	7	54499.5644	7	54879.7384	18	55875.8397	26
52775.3237	4	53890.6318	7	54501.37596	WASP	54903.6043	7	55890.9069	26
52776.2882	4	53981.8799	7	54501.49785	WASP	54903.7252	7	55921.7637	26
52785.6871	7	54009.8443	7	54501.61833	WASP	54910.4749	17	55921.8832	26
52983.4844	8	54015.8703	7	54502.46160	WASP	54911.4396	17	55932.6119	23
53011.5660	7	54058.7811	7	54502.58141	WASP	54912.4054	17	55932.7321	23
53028.4459	8	54067.5800	12	54504.38969	WASP	54912.5263	19	55935.6269	23
53051.5825	7	54067.7015	12	54506.6801	7	54923.6165	7	55936.5888	23
53052.4285	9	54079.8751	7	54511.5016	7	54924.3380	17	55958.5270	23
53064.72252	6	54085.2999	12	54511.6224	7	54943.3822	17	55958.6474	23
53070.3860	9	54085.4205	12	54511.7420	7	55114.9109	7	55969.6165	23
53079.6693	7	54085.5652[†]	12	54512.4658	14	55163.8484	7	55995.6533	23
53090.3992	9	54091.3232	12	54513.6690	7	55167.7045	7	55999.6306	23
53094.3780	9	54091.4482	12	54520.7815	7	55167.8271	7	56001.3182	27
53096.4229	9	54091.5392[†]	12	54524.39764	WASP	55167.9454	7	56002.5231	15
53099.4361	9	54116.2770	12	54524.51953	WASP	55189.5234	19	56008.4330[†]	15
53119.3258	10	54116.3962	12	54525.36216	WASP	55191.4519	19	56008.5510	15
53125.3520	10	54116.5168	12	54525.6041	7	55191.5722	19	56012.4070	15
53129.3303	10	54120.8567	7	54526.44753	WASP	55191.6931	19	56012.5320[†]	15
53132.3426	10	54134.7170	7	54527.53236	WASP	55206.7590	18	56329.30423	XL
53145.3601	10	54168.5878	7	54530.42492	WASP	55209.6519	18		
53338.81318	6	54173.4080	13	54530.54605	WASP	55210.7380	18		
53395.58528	6	54180.4001	13	54532.35361	WASP	55212.6651	18		

[*] References: (1) Hipparcos; (2) Agerer and Hübscher 2002; (3) Derekas et al. 2003; (4) Gazeas, Niarchos, and Boutsia 2004; (5) Agerer and Hübscher 2003; (6) Klingenberg, Dvorak, and Robertson 2006; (7); Samolyk 2010; (8) Hübscher, Paschke, and Walter 2005; (9) Hübscher 2005; (10) Gazeas and Niarchos 2005; (11) Hübscher, Paschke, and Walter 2006; (12) Hübscher and Walter 2007; (13) Hübscher 2007; (14) Hübscher, Steinbach, and Walter 2009a; (15) Hübscher, Braune, and Lehmann 2013; (16) Hübscher, Steinbach, and Walter 2009b; (17) Hübscher et al. 2010; (18) Samolyk 2011; (19) Wils et al. 2010; (20) Hübscher and Lehmann 2012; (21) Hübscher and Monninger 2011; (22) Wils et al. 2011; (23) Samolyk 2013; (24) Wils et al. 2012; (25) Hübscher 2011; (26) Samolyk 2012; (27) Wils et al. 2013. WASP: WASP project; YN: Yunnan observatory; XL: Xinglong observation station.

[†] Not used in analysis because of large discrepancy with the other data.

δ Scuti type stars can be checked by using data observed by the WASP project.

The Hipparcos Intermediate Astrometric Data (HIAD) are one-dimensional coordinates (abscissae) on reference great circles obtained, by an independent analysis, but using the same basic satellite data, by the FAST and NDAC Data Reduction Consortia (Perryman et al.1997). Recently, the HIAD were revised, and new data were given (van Leeuwen 2007). Combining with spectroscopic observations or photometric observations, they have been used to obtain orbital solutions of many binaries (Ren & Fu 2010 and the references there in). The parallax of SZ Lyncis was estimeated to be 2.23 ± 1.16 mas (Perryman et al. 1997), which corresponds to a distance of 448 pc. However, a recent reassessment of the Hipparcos astrometry by van Leeuwen (2007) derived 1.00 ± 1.12 mas, placing it at a distance of 1000 pc, which disagrees with the finding reported by Bardin and Imbert (1984).

In order to investigate the pulsating and orbital elements of the SZ Lyncis system, we determined dozens of new times of the light maximum from our observations and the WASP project, collected all of the times of the light maximum that have been published, and studied the behavior of the orbit of SZ Lyncis system using the classical $O-C$ method. We also used the HIAD to obtain the complete orbital elements of the SZ Lyncis system. This paper reports on photometric observation in section 2 and analyzes the $O-C$ diagram and astrometric solution in sections 3 and 4. Sections 5 and 6 present discussions and conclusions, respectively.

2. Observations

Our observations were carried out on 2008 November 10 and 2013 February 05, using the 85 cm telescope at the Xinglong observation station of the National Astronomical Observatories of the Chinese Academy of Sciences (Zhou et al. 2009), and 2008 April 21, using the 1 m telescope at Yunnan Observatory. The exposure times ranged from 3 to 40 s, depending on the diameter of the telescopes, filters and nightly seeing. All observations were reduced using standard IRAF procedures. GSC02979-01329 (08:09:55.8 +44:25:48.67, $V = 10.8$) was selected as a comparison star, and the check stars were GSC02979-01343 (08:09:29.8 +44:24:28.8, $V = 10.8$) and GSC02979-01370 (08:09:38.8 +44:26:46.2, $V = 13.3$). Three new times of the light maximum were determined by fitting a third-degree polynomial on each observed peak (see table 1).

SZ Lyncis was observed by the WASP project between 2007 March 30 and 2008 April 18, during which 2894 data points were obtained in the observations. The telescope has eight 20 cm lenses, each of which covers $7^{\circ}\!.8 \times 7^{\circ}\!.8$. A 2048×2048 thinned CCD with a pixel size of 13.5 μm is located behind each lens (Butters et al. 2010). The exposure time was fixed as 30 s. The light curves were given by the TAMUZ corrected flux (Collier Cameron et al. 2006), which could be converted to magnitude by using mag = 15−2.5 log(flux). Based on the observations, thirty-nine new times of light maximum were determined by us, and the method was the same as mentioned above (see table 1).

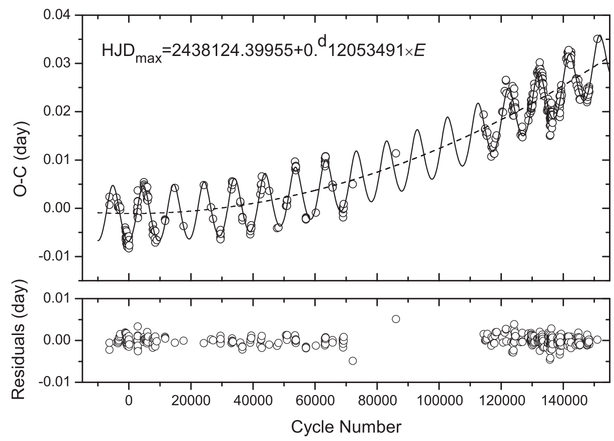

Fig. 1. $O-C$ diagram of SZ Lyncis from the linear ephemeris of equation (1). All times of light maximum have been used. The solid line in the top panel shows the fit containing an increasing pulsation period change and the light-travel time effect component. The residuals are presented in the lower panel.

3. Analysis of $O-C$ Diagram

The $O-C$ analyses of SZ Lyncis have been conducted by many authors (Paparó et al. 1988; Gazeas et al. 2004; Gazeas & Niarchos 2005). Based on their work, 393 times of light maximum observed between 1961 and 2013 were used for our $O-C$ analysis. Some data with an unusually large error were not used in the analysis. The linear ephemeris that we used is given by Paparó et al. (1988):

$$\mathrm{HJD_{max}} = 2438124.39955 + 0^{\mathrm{d}}\!.12053491 \times E. \tag{1}$$

The corresponding $O-C$ diagram is displayed in figure 1. To describe the $O-C$ curve well, the combination of a quadratic ephemeris and an additional periodic terms are required:

$$O-C = \delta T_0 + \delta P_0 E + \frac{\beta}{2} E^2 + \tau, \tag{2}$$

where δT_0 and δP_0 are the correction values of the epoch and the pulsation period in equation (1), β is the linear change of the pulsation period (d cycle^{-1}), and τ is the cyclic change caused by the light-travel time effect,

$$\begin{aligned}\tau &= A\left[(1-e^2)\frac{\sin(\nu+\omega)}{1+e\cos\nu} + e\sin\omega\right] \\ &= A\left[\sqrt{1-e^2}\sin E^*\cos\omega + \cos E^*\sin\omega\right].\end{aligned} \tag{3}$$

Here, $A = (a_1 \sin i)/c$ is the projected semi-major axis given in day, e is the eccentricity, ν is the true anomaly, ω is the longitude of the periastron passage in the plane of the orbit, and E^* is the eccentric anomaly. This equation was first given by Irwin (1952a).

The Kepler equation gives the connection between E^* and the mean anomaly, M:

$$M = E^* - e\sin E^* \tag{4}$$

and

$$M = \frac{2\pi}{P_{\mathrm{A}}}(t-T). \tag{5}$$

Table 2. Pulsating and orbital elements of the SZ Lyncis system.

Parameter	All 1961–2013	New 2000–2013
T_0[cor]	$2438124.39846 \pm 0.00016$	52776.28105 ± 0.00021
P_0[cor]	$0\overset{\mathrm{d}}{.}120534908 \pm 0.000000005$	$0\overset{\mathrm{d}}{.}120535208 \pm 0.000000023$
β (d cycle^{-1})	$(2.74 \pm 0.07) \times 10^{-12}$	2.74×10^{-12} (fixed)
A (d)	0.00577 ± 0.00008	0.00589 ± 0.00012
$a_1 \sin i$ (au)	1.000 ± 0.015	1.019 ± 0.021
e	0.16 ± 0.03	0.20 ± 0.04
ω (°)	78.2 ± 11.0	67.3 ± 12.1
P_{orb} (d)	1181.8 ± 0.5	1185.8 ± 4.7
T	2445738.8 ± 35.7	2445672.0 ± 40.2

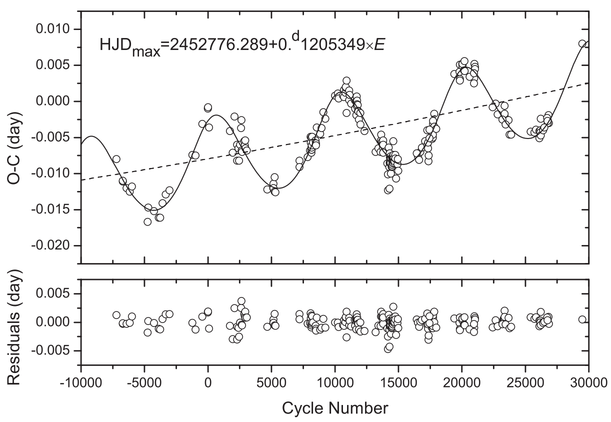

Fig. 2. $O - C$ diagram of SZ Lyncis from the linear ephemeris of equation (6). Only the times of light maximum obtained between 2000 and 2013 have been used.

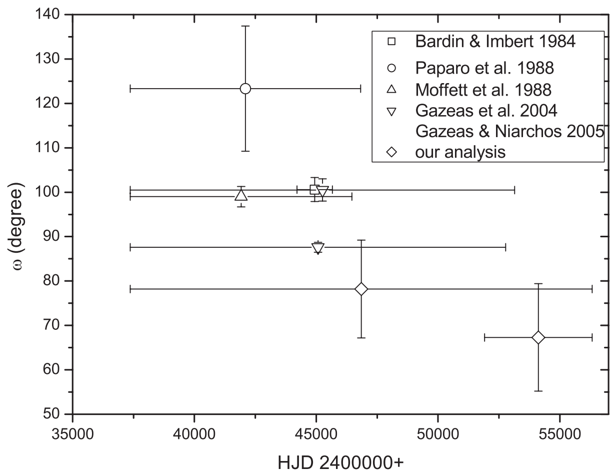

Fig. 3. Longitude of the periastron passage of the SZ Lyncis system given by early literature and our analysis. The abscissa is the mean time of the observations, and the error bar along the abscissa indicates the range of data used in the determination.

Here, P_{A} is the anomalistic period, defined by the interval of time between two consecutive periastron passages; here, it is the orbital period, t is the time of light maximum and T is the time of passage through the periastron. The results of a least-squares solution by the Levenberg–Marquardt (LM) method are given in column 2 of table 2. Figure 1 shows an $O - C$ diagram fitted with the parameters. The longitude of the determined periastron passage is $78\overset{\circ}{.}2 \pm 11\overset{\circ}{.}0$, lower than the values given by the early literature.

After the work of Paparó, Szeidl, and Mahdy (1988), SZ Lyncis was neglected until the 21st Century. In order to find the latest orbital parameters of SZ Lyncis, we selected 240 times of the light maximum between 2000 and 2013 for an $O - C$ analysis. The linear ephemeris is given by Gazeas et al. (2004):

$$\mathrm{HJD}_{\max} = 2452776.289 + 0\overset{\mathrm{d}}{.}1205349 \times E. \quad (6)$$

In the analysis, because of the shorter span of the times of the light maximum, we fixed value of β as a constant, 2.74×10^{-12} d cycle^{-1}. It was obtained from an early fitting, in which all of the times of the light maximum spanning about 50 yr were used. The results are given in column 3 of table 2. Figure 2 shows the $O - C$ diagram. The longitude of the periastron passage, ω, determined is $67\overset{\circ}{.}3 \pm 12\overset{\circ}{.}1$. Published values of ω determined by different methods are plotted in figure 3. It seems that ω is decreasing with time.

In order to measure the change of ω, we ignore the short-period oscillation, and consider that ω changes linearly with time (Morais & Correia 2012),

$$\omega = \omega_0 + \dot{\omega} E, \quad (7)$$

where ω_0 is the longitude of the periastron at the initial time and $\dot{\omega}$ is the angular velocity of the line of apsides. In eclipsing binaries with apsidal motion, there is a relationship between the sidereal and anomalistic periods (Bozkurt & Değirmenci 2007),

$$P_{\mathrm{S}} = P_{\mathrm{A}} \left(1 - \frac{\dot{\omega}}{2\pi}\right), \quad (8)$$

where the units of $\dot{\omega}$ here are radians per sidereal period. The value of $\dot{\omega}$ is estimated by fitting a precessing Keplerian orbit [equations (2), (3), and (7)] to the $O - C$ data. The corresponding results of the fit by the Simplex method are listed in Table 3. The value of $\dot{\omega}$ is $(-1\overset{\circ}{.}15 \pm 0\overset{\circ}{.}25)$ yr^{-1}. It should be mentioned that the LM method is also used in the fit, but the results are not single. For the sake of checking the reliability of the results, we set $\dot{\omega}$ as being fixed values that change with

Table 3. Fitted parameters for the SZ Lyncis system.*

Parameter	
T_0 [cor]	$2438124.39849 \pm 0.00018$
P_0 [cor]	$0^{\mathrm{d}}.120534908 \pm 0.000000005$
β (d cycle^{-1})	$(2.73 \pm 0.07) \times 10^{-12}$
A (d)	0.00579 ± 0.00021
$a_1 \sin i$ (au)	1.002 ± 0.037
e	0.17 ± 0.03
ω_0 (°)	117.6 ± 12.1
$\dot{\omega}$ (° yr^{-1})	-1.15 ± 0.25
P_{A} (d)	1169.9 ± 2.3
P_{S} (d)	1182.1 ± 2.3
T	2445786.4 ± 21.3

* It is considered that parameter ω changes linearly with time, and equations (2), (3), and (7) are used in the fit.

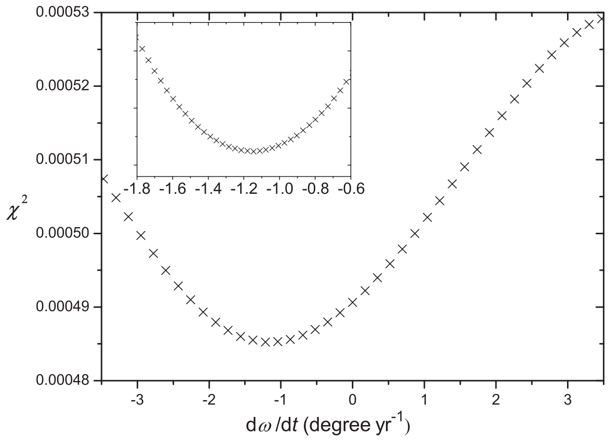

Fig. 4. Reduced χ^2 as a function of $\dot{\omega}$. The minimal value of χ^2 was obtained near $-1^{\circ}.2\,\mathrm{yr}^{-1}$, and the details can be clearly seen in the inset.

small steps, and obtained the reduced χ^2 as a function of $\dot{\omega}$. As can be seen in figure 4, the best fit with the LM optimization algorithm is obtained near a value of $-1^{\circ}.2\,\mathrm{yr}^{-1}$, which agrees with the results given by the Simplex Method.

4. Hipparcos Observations and Astrometric Solution

The SZ Lyncis system was observed by the Hipparcos satellite during its 37-month astrometric mission, and was measured a total of forty-eight times from 1990 March to 1992 May, slightly over two-thirds of the orbital cycles of the binary. Each measurement consisted of a one-dimensional position ("abscissa," υ) along a great circle representing the scanning direction of the satellite, tied to the absolute frame of reference, known as the International Celestial Reference System. The measurements available were published in the form of "abscissa residuals" ($\Delta\upsilon$), which were the residuals from the standard five-parameter solution reported in the catalog (Perryman et al. 1997). The five standard parameters are the position (α_0^*, δ_0) and the proper-motion components (μ_α^*, μ_δ) of the barycenter at the reference epoch, 1991.25, and the trigonometric parallax, π_t. The formalism for modeling the abscissa residuals closely follows that described by van Leeuwen and Evans (1998), Pourbaix and Jorissen (2000), and Jancart et al. (2005), including the correlations between measurements from the two independent data-reduction consortia (Northern Data Analysis Consortium, NDAC and Fundamental Astrometry by Space Techniques, FAST: Perryman et al. 1997).

In the most general case, the adjustable quantities of the fit are the corrections to the five standard Hipparcos parameters ($\Delta\alpha^*$, $\Delta\delta$, $\Delta\mu_\alpha^*$, $\Delta\mu_\delta$, and $\Delta\pi_t$), and seven usual elements of the binary orbit: the orbital period, P, the semimajor axis of the photocenter, a_{phot}, the eccentricity, e, the inclination angle, i, the longitude of periastron for the primary, ω_A, the position angle of the ascending node, Ω (equinox J2000.0), and the time of periastron passage, T. For SZ Lyncis the Hipparcos measurements do not constrain P, e, ω_A, or T, so those elements were held fixed at their values, as given by our $O-C$ analysis.[1] The remaining eight parameters were fitted using standard least-squares techniques. In the analysis, we found that the orbital inclinations, i, is unrealistically close to zero. Perhaps the parallax (2.23 mas) of SZ Lyncis is too small to derive reliable results, and some restrictions should been made.

Because SZ Lyncis is reported to be a single-lined spectroscopic binary, we assume that there is no light coming from the secondary (Jancart et al. 2005). Hence, the orbit of the photocenter of the primary component is the same as the absolute orbit of the primary around the center of the mass of the system,

$$a_{\mathrm{phot}} = a_{\mathrm{rel}} \frac{q}{1+q} = (a_1 \sin i)(\pi_0 + \Delta\pi_t)/\sin i, \tag{9}$$

where $q \equiv M_{\mathrm{B}}/M_{\mathrm{A}}$, a_{rel} is the relative semimajor axis; a_{phot} is related to the parameters $\Delta\pi_t$ and i, and there are only seven parameters in the fit. The results are listed in column 2 of table 4.

An F-test is used to check whether the orbital solution represents a significant improvement over the 5-parameter Hipparcos solution. In the test, the false-alarm rate of rejecting the null hypothesis can be expressed as (Pourbaix 2001)

$$Pr_1 = Pr[F(2, N-7) > \hat{F}], \tag{10}$$

where

$$\hat{F} = \frac{N-7}{2} \frac{\chi^2_{\mathrm{single}} - \chi^2_{\mathrm{orb}}}{\chi^2_{\mathrm{orb}}} \tag{11}$$

follows a F-distribution with (2,$N-7$) degrees of freedom and N is the number of available data; χ^2_{single} and χ^2_{orb} are the χ^2 values associated with the single-star and orbital models, respectively. The correlation among the derived parameters is indicated by using the concept of *efficiency*, ϵ (Eichhorn 1989; Jancart et al. 2005), which is defined as

$$\epsilon = \sqrt[p]{\frac{\prod_{k=1}^{p} \lambda_k}{\prod_{k=1}^{p} q_{kk}}}, \tag{12}$$

where λ_k are the eigenvalues of the covariance matrix of the estimated parameters, q_{kk} are its diagonal elements, and

[1] Considering the parameter ω may change with time, using the equation (7), the value of ω obtained is 85° at 1991.25. This value is used in the solution.

Table 4. Astrometric orbital solutions for the SZ Lyncis system.

Parameter	Old data Perryman et al. (1997)	Revised van Leeuwen (2007)
$a_{\rm phot}$ (mas)	4.12 ± 1.66	3.26 ± 3.05
i (°)	39.5 ± 17.7	23.9 ± 22.8
Ω (°)	125.5 ± 12.4	118.5 ± 18.2
$\Delta\alpha^*$ (mas)	1.18 ± 0.96	0.98 ± 1.09
$\Delta\delta$ (mas)	-1.18 ± 0.71	-0.90 ± 0.74
$\Delta\mu_\alpha^*$ (mas yr^{-1})	-0.23 ± 1.62	-0.21 ± 1.73
$\Delta\mu_\delta$ (mas yr^{-1})	-3.63 ± 1.62	-4.19 ± 1.64
$\Delta\pi_t$ (mas)	0.38 ± 0.98	0.34 ± 1.24
μ_α^* (mas yr^{-1})	-9.95 ± 1.62	-10.98 ± 1.73
μ_δ (mas yr^{-1})	-29.66 ± 1.62	-30.75 ± 1.64
π_t (mas)	2.61 ± 0.98	1.32 ± 1.24
Distance (pc)	380 ± 140	760 ± 720
$\chi^2_{\rm orb}$	31.8495	112.0208
$\chi^2_{\rm single}$	38.4313	120.2305
Pr_1	2.1%	4.6%
ϵ	0.58	0.53

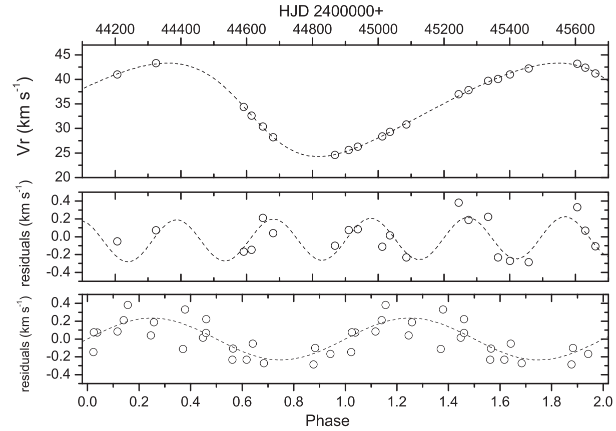

Fig. 5. Top: twenty-one mean radial-velocity values given by Bardin and Imbert (1984). The dash line is a single-Keplerian fit. Middle: residual velocities from the Keplerian fit showing periodic or quasi-periodic change. The dash line is the fit to the data with a sinusoid ($K = 0.235\,{\rm km\,s^{-1}}$, $P = 295\,{\rm d}$). Bottom: residual velocities as the function of phase.

$p = 7$ denotes the number of parameters in the model. In general, when $Pr_1 < 5\%$, and $\epsilon \geq 0.4$, the orbital solution may be accepted. The values of Pr_1 and ϵ in our solution are 2.1% and 0.58, respectively, which means that our solution is accepted. The revised HIAD given by van Leeuwen (2007) were also used in the analysis. However, the errors of the results are much larger, and the smaller parallax is not in accord with the result obtained by the other method (Bardin & Imbert 1984). The parameters derived from the old HIAD are more reliable.

5. Discussion

The linear changes of the longitude of the periastron passage in a binary system, also called apsidal motion, have been observed in many detached eclipsing binaries and binary pulsars (Petrova & Orlov 1999; Bulut & Demircan 2007 and their references). These were consider to be caused by several different effects, such as the general-relativistic effect, tidel distortion, rotational flattening, the third component, or combined effects (Claret 1997; Quataert et al. 1996; Claret & Giménez 1993; Bozkurt & Değirmenci 2007). For the SZ Lyncis system, due to the orbital period and the relative semimajor axis, the contribution of these effects can not explain the large rate of the precession. Recently, Morais and Correia (2012) modeled the secular evolution of a star's orbit when it has a nearby binary system, and showed that the major secular effect was precession of the star's orbit around the binary system's centre of mass. They analyzed the radial-velocity data for a single-line K-giant binary ν Octantis ($P \simeq 2.9\,{\rm yr}$, $e = 0.2358$, Ramm et al. 2009), and obtained a retrograde precession of $(-0^\circ\!.86 \pm 0^\circ\!.02)\ {\rm yr}^{-1}$. Considering SZ Lyncis having similar orbital parameters and precession rate as that of ν Octantis, the same mechanism probably reflects the orbit precession of SZ Lyncis, which means that the companion of SZ Lyncis should be a close binary system.

The inclination, i, obtained by us is $39^\circ\!.5 \pm 17^\circ\!.7$, which disagrees with the conclusion of Moffett et al. (1988), who claimed that the orbital inclination, i, is probably greater than 47°. However, their points are based on the fact that SZ Lyncis is a single-lined spectroscopic binary, which means that the magnitude difference between the two components should be at least 1.5 mag (Paparó et al. 1988), and the assumption that the companion should be a single star. Fernley et al. (1984) concluded that the mass and absolute magnitude, Mv, of SZ Lyncis should be $1.57\,M_\odot$ and 1.71 mag. Using the mass function, the obtained mass of the companion, $M_{\rm B}$, is about $1.5\,M_\odot$. If it was a single main sequence star, its spectral type should be F2 (corresponding to an absolute magnitude, Mv, of ~ 3.6 mag, Drilling & Landolt 2000). This is about 1.9 mag fainter than the primary star. If the companion was a binary system, the combined luminosity would be lower, and the high inclination is no longer necessary.

Morais and Correia (2012) also pointed out that if ν Octantis B was, itself, a double star, it could mimic a signal with similarities to that previously identified as a planet of ν Octantis A. A similar phenomenon may exist in the SZ Lyncis system. However, no periodic behaviour in the residuals of the $O - C$ values was found in our analysis, but a weak variation was discovered in the residuals of the mean radial velocities from a single-Keplerian fit (Irwin 1952b). A sine function was used for fitting the data, and a period of 295 d and a semi-amplitude of $235\,{\rm m\,s^{-1}}$ were derived (see figure 5). It is surprising that the period is nearly commensurate with the orbital period of the SZ Lyncis system (ratio 1:4). Noticing that the mean radial-velocity curve is accurate to $200\,{\rm m\,s^{-1}}$, more detailed high-precision spectroscopic observations are definitely needed for checking the existance of a shorter periodic change.

6. Conclusion

Forty-two new times of the light maximum were determined from our photometry observations and the WASP project. All of the times of the light maximum observed between 1961 and 2013 were used to calculate the orbital elements of the SZ Lyncis system and the secular change of the pulsation period with the classical $O-C$ method. The decrease in the longitude of the periastron passage, ω, with a rate of $(-1\overset{\circ}{.}15 \pm 0\overset{\circ}{.}25)\,\mathrm{yr}^{-1}$ was confirmed. The causative mechanisms were discussed, and the most likely explanation was that the precession of the star's orbit was due to a close binary system. This meant that the companion of SZ Lyncis should be a close binary system. The Hipparcos Intermediate Astrometric Data were used to obtain complete orbital elements of the SZ Lyncis system. The inclination, i, obtained was $(39\overset{\circ}{.}5 \pm 17\overset{\circ}{.}7)$, which is not in agreement with the previous work. However, the fact that high inclination was not necessary when the companion was a close binary system was pointed out. The revised parallax, π_t, was 2.61 ± 0.98, which corresponds to $380 \pm 140\,\mathrm{pc}$. Even though the error was big, the result agreed with the result given by Bardin and Imbert (1984). After a reanalysis of the mean radial velocities of SZ Lyncis given by Bardin and Imbert (1984), a weak signal with period of 295 d was discovered in the residuals from fitting a single-Keplerian orbit. It was also noticed that the period is nearly commensurate with the orbital period of the SZ Lyncis system (ratio 1:4). To check the existance of this shorter periodic change, more high-precision spectroscopic observations are definitely needed in the future.

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